```
num = int(input("Enter an Integer : "))
if num < 0:
  print("Negative number!")
else:
  print("Positive Number!")
```

```
Enter an Integer : -20
Negative number!
```

Riya Kathe
11:09 AM Today

1. Write a program to check whether a given number is positive or negative.

```
num=int(input("Enter a  number"))
if num%2==0:
  print(num,"Even")
else :
  print(num,"Odd")
```

```
Enter a  number78
78 Even
```

Riya Kathe
11:12 AM Today

2. Write a program to check whether a number is even or odd.

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are eligible to vote.")
else:
    print("You are not eligible to vote.")
```

```
Enter your age: 87
You are eligible to vote.
```

Riya Kathe
11:14 AM Today

3. Write a program to check whether a person is eligible to vote or not. Take age as input.

```
num1 = int(input("enter a number"))
num2 = int(input("enter a number"))

if num1>num2:
  print(num1,"is greater than",num2);
else:
  print(num2,"is greater than",num1)
```

```
enter a number20
enter a number60
60 is greater than 20
```

Riya Kathe
11:10 AM Today

4. Write a program to find the greater number between two numbers entered by the user.

```
marks = int(input("Enter student marks: ")

if marks >= 40:
    print("Student has passed.")
else:
    print("Student has failed.")
```

```
Enter student marks: 90
Student has passed.
```

Riya Kathe
11:15 AM Today
(edited 11:15 AM Today)

5. Write a program to check whether a student has passed or failed. Passing marks are 40.

```
number=int(input("enter the number: "))
if number%5==0 &number%10==0:
  print("number is divisible")
```

Riya Kathe
11:16 AM Today

```
else:
  print("number is not divisible")
```

```
enter the number: 50
number is divisible
```

6. Write a program to check whether a number is divisible by both 5 and 10.

```
s = input('Enter a character : ').lower()
if s in "aeiou" :
  print('Character is a vowel..!')
else:
  print('Character is a constant')
```

```
Enter a character : B
Character is a constant
```

Riya Kathe
11:17 AM Today

7. Write a program to check whether a given character is a vowel or a consonant.

```
n=int(input("Enter Electricity Bill "))
print("Electricity bill",n*7)
```

```
Enter Electricity Bill 10
Electricity bill 70
```

Riya Kathe
11:17 AM Today

8. Write a program to calculate electricity bill based on units consumed.

```
year = int(input("Enter a year: "))

if (year % 400 == 0) or (year % 4 == 0 and
    print(year, "is a leap year")
else:
    print(year, "is not a leap year")
```

```
Enter a year: 2005
2005 is not a leap year
```

Riya Kathe
11:17 AM Today

9. Write a program to check whether a year is a leap year or not.

```
salary = float(input("Enter employee salar

if salary > 50000:
    bonus = salary * 0.10
elif salary >= 30000:
    bonus = salary * 0.07
else:
    bonus = salary * 0.05

print("Bonus for the employee is:", bonus)
```

```
Enter employee salary: 5000
Bonus for the employee is: 250.0
```

Riya Kathe
11:18 AM Today

10. Write a program to calculate bonus for an employee based on salary.

```
for i in range (1,11):
 print(i)
```

```
1
2
3
4
5
6
7
8
```

Riya Kathe
11:18 AM Today

11. Write a program to print numbers from 1 to 10 using a loop.

```
9
10
```

```
for i in range (1,51):
 if i%2==0 :
  print(i)
```

```
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

Riya Kathe
11:18 AM Today

12. Write a program to print all even numbers between 1 and 50.

```
sum = 0
for i in range(1, 11):
    sum += i

print("Sum of first 10 natural numbers is:
```

```
Sum of first 10 natural numbers is: 55
```

Riya Kathe
11:19 AM Today

13. Write a program to calculate the sum of first 10 natural numbers.

```
num=int(input("Enter a Number for multipli
for i in range(1,11):
    print(num*i)
```

```
Enter a Number for multiplication table2
2
4
6
8
10
12
14
16
18
20
```

Riya Kathe
11:19 AM Today

14. Write a program to print the multiplication table of a number entered by the user.

Riya Kathe
11:20 AM Today

```
num = int(input("Enter a number: "))
count = 0

while num > 0:
    num = num // 10   # remove last digit
    count += 1

print("Number of digits:", count)
```

```
Enter a number: 20
Number of digits: 2
```

15. Write a program to count the number of digits in a given number.

Riya Kathe
11:20 AM Today

16. Write a program to check pass or fail for 5 students using loop and condition.

```
for i in range (1,6):
  stud_marks=int(input("Enter a marks"))
  if stud_marks>=40 :
    print("pass")
  else :
    print("fail")
```

```
Enter a marks45
pass
Enter a marks89
pass
Enter a marks78
pass
Enter a marks565
pass
Enter a marks65
pass
```

Riya Kathe
11:20 AM Today

17. Write a program to print numbers from 1 to 100 and display Even or Odd for each.

```
for i in range (1,101):
 if i%2==0:
  print("Even",i)
 else :
  print("Odd",i)
```

```
Odd 1
Even 2
Odd 3
Even 4
Odd 5
Even 6
Odd 7
Even 8
Odd 9
Even 10
Odd 11
Even 12
Odd 13
Even 14
Odd 15
Even 16
Odd 17
Even 18
Odd 19
Even 20
Odd 21
Even 22
```

```
Odd 23
Even 24
Odd 25
Even 26
Odd 27
Even 28
Odd 29
Even 30
Odd 31
Even 32
Odd 33
Even 34
Odd 35
Even 36
Odd 37
Even 38
Odd 39
Even 40
Odd 41
Even 42
Odd 43
Even 44
Odd 45
Even 46
Odd 47
Even 48
Odd 49
Even 50
Odd 51
Even 52
Odd 53
Even 54
Odd 55
Even 56
Odd 57
```

Riya Kathe
11:20 AM Today

18. Write a program to find how many numbers between 1 and 50 are divisible by 3.

```
for i in range(1,51):
  if i%3==0 :
    print(i,"Is divisible by 3")
```

```
3 Is divisible by 3
6 Is divisible by 3
9 Is divisible by 3
12 Is divisible by 3
15 Is divisible by 3
18 Is divisible by 3
21 Is divisible by 3
24 Is divisible by 3
27 Is divisible by 3
30 Is divisible by 3
33 Is divisible by 3
36 Is divisible by 3
39 Is divisible by 3
42 Is divisible by 3
45 Is divisible by 3
48 Is divisible by 3
```

Riya Kathe
11:21 AM Today

19. Write a program to take marks of 5 students and calculate the average.

```
total = 0

for i in range(5):
    marks = int(input("Enter marks of stud
    total += marks
```

```
average = total / 5
print("Average marks of 5 students is:", a
```

```
Enter marks of student : 90
Enter marks of student : 80
Enter marks of student : 60
Enter marks of student : 60
Enter marks of student : 43
Average marks of 5 students is: 66.6
```

Riya Kathe
11:22 AM Today

20. Write a program to take 5 numbers from the user and count positive and negative numbers.

```
positive_count = 0
negative_count = 0

for i in range(5):
    num = int(input(f"Enter number {i+1}: "))
    if num > 0:
        positive_count += 1
    elif num < 0:
        negative_count += 1

print("Positive numbers:", positive_count)
print("Negative numbers:", negative_count)
```